SEISAGENTBENCH: BENCHMARKING AND REINFORCEMENT LEARNING FOR SEISMIC DATA-PROCESSING AGENTS

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ABSTRACT

Seismic data processing is a realistic scientific workflow in which experts assemble and revise multi-stage pipelines over wavefield data. Unlike isolated prediction tasks, success depends on a sequence of data-dependent choices and on the scientific quality of the resulting artifact. Existing agent benchmarks seldom combine executable, multi-turn interaction with task-specific evaluation of scientific data products. We introduce **SeisAgentBench**, to our knowledge the first executable benchmark designed for language agents to construct and evaluate seismic artifacts through multi-step processing, together with **SeisAgent**, an LLM policy specialized for this setting through trajectory-level supervised learning and group-relative reinforcement-learning post-training. SeisAgentBench contains 95 tasks across 19 processing families and evaluates outputs using deterministic, family-specific seismic diagnostics, compatible reference agreement, artifact-preservation checks, and workflow cost. Together, SeisAgentBench and SeisAgent provide a controlled testbed for planning, numerical parameter selection, feedback-driven revision, and scientific artifact evaluation.

1 INTRODUCTION

Large language models (LLMs) have enabled agents that translate natural-language goals into multi-step actions, interact with external software, and adapt to execution feedback (Achiam et al., 2023; Yao et al., 2022a; Jimenez et al., 2024; Yao et al., 2022b; Schick et al., 2023). Agent benchmarks now test stateful interaction in web environments, software repositories, shopping systems, and algorithm-development settings (Lu et al., 2025; Wang et al., 2026; Sun et al., 2026). These benchmarks have made end-to-end execution a central evaluation target. However, scientific data processing raises a further requirement: *the final object may be a valid file produced by a legal sequence of actions but still be scientifically unusable* (Chen et al., 2025; Ai et al., 2026). A blanked image, an over-muted gather, an implausible physical model, or a structurally damaged section cannot be judged from completion alone.

Seismic data processing is a demanding case of this problem. It transforms recorded wavefields into products used for subsurface analysis and imaging (Yilmaz, 2001). A workflow may combine noise attenuation, velocity analysis, moveout correction, stacking, migration, and model construction (Dix, 1955; Robinson, 1957). These operations are coupled: aggressive filtering can remove useful reflections, while an inaccurate velocity field can damage later stacking and imaging. Suitable parameters are rarely known in advance, and several choices can appear reasonable under noisy observations. Experts therefore inspect intermediate products, compare alternatives, revise earlier decisions, and stop when further processing is more likely to damage than improve the result (Yilmaz, 2001). Learning-based seismic methods have produced strong results for individual operations such as denoising, velocity estimation, inversion, and imaging (Yu & Ma, 2021; Khosro Anjom et al., 2024). Their usual formulation is a fixed mapping for one task and one output type. Recent LLM systems have also supported geoscience data exploration and seismic workflow assistance (Widlansky & Komar, 2025; Kanfar et al., 2025). These systems support exploration or workflow assistance, but do not provide a shared multi-family benchmark that evaluates both executable processing trajectories and the resulting seismic artifacts. What remains missing is a **shared executable setting** in which one policy must work across heterogeneous seismic products, select parameterized operations, use intermediate scientific evidence, revise candidates, and return a scientifically supported artifact.

We address this gap with **SeisAgentBench**, an executable benchmark of 95 tasks across 19 seismic-processing families. As shown in Figure 1, each episode produces a processing trajectory and a

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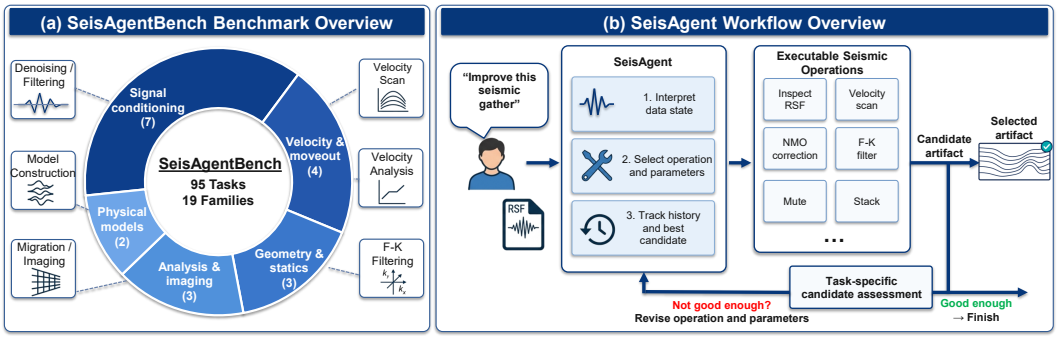


Figure 1: **Overview of SeisAgentBench and SeisAgent.** (a) SeisAgentBench contains 95 tasks across 19 seismic-processing families, grouped by scientific objective and output type. (b) SeisAgent interprets the artifact state, selects a parameterized operation, and receives task-specific candidate evidence. Weak candidates trigger revision, while a supported candidate terminates the episode. RSF, NMO and F–K denote regularly sampled format, normal moveout and frequency–wavenumber, respectively.

seismic artifact whose lineage and scientific properties can be examined together. A persistent workspace supports candidate comparison and revision without discarding earlier products, while domain-specific evaluation checks the intended processing effect, signal preservation, and interaction cost. We also introduce **SeisAgent**, which combines a small trajectory-level supervised fine-tuning (SFT) warm start (Ouyang et al., 2022) with group-relative reinforcement learning (Group-RL) (Shao et al., 2024). Comparing alternative workflows for the same task provides a common basis for learning operation selection, numerical parameters, revision, and stopping across heterogeneous seismic products.

In summary, we advance seismic-agent research in two connected directions. First, we introduce SeisAgentBench, to our knowledge **the first shared executable benchmark for language-agent seismic processing across multiple families**, combining artifact lineage, scientific diagnostics, preservation checks, and workflow evaluation. Second, we introduce SeisAgent, which learns processing decisions through trajectory supervision and scientific-feedback Group-RL. Together, they establish a common setting for evaluating and training agents to construct, assess, and revise seismic products, moving beyond isolated prediction tasks and individual operation calls.

2 Preliminaries

In this section, we introduce the seismic concepts needed to understand the benchmark and then formalize agentic seismic processing.

2.1 Seismic Data Processing

Seismic surveys record wavefield amplitudes as collections of traces, with each trace sampling amplitude over time for a source–receiver configuration. Coherent arrivals that can be tracked across neighboring traces are commonly called *seismic events*. Across processing stages, workflows operate on and produce shot gathers, common-midpoint (CMP) gathers, stacked sections, migrated images, velocity models, and other scientific products (Yilmaz, 2001; Mayne, 1962). SeisAgentBench is designed around this range of intermediate and final products.

A conventional linear workflow transforms an initial artifact x_0 through parameterized operations

$$x_k = f_k(x_{k-1}; \phi_k), \qquad k = 1, \ldots, K. \tag{1}$$

Here, x_0 may be a gather, spectrum, image, physical model, or a related collection of arrays and metadata, depending on the task. SeisAgentBench generalizes this linear form: an operation may consume one or more artifacts from the current workspace, and its output is added to a persistent lineage graph rather than replacing the previous state. A poor parameter choice may remove useful signal, introduce artifacts, or reduce the value of later processing. Suitable parameters must be inferred from noisy data and intermediate results, often with several plausible alternatives. Seismic processing therefore depends on repeated inspection and task-specific quality control (QC) rather than software execution alone.

Figure 2 gives a representative normal moveout (NMO) example (Taner & Koehler, 1969; Neidell & Taner, 1971). The processor estimates a velocity field, applies correction, and checks whether

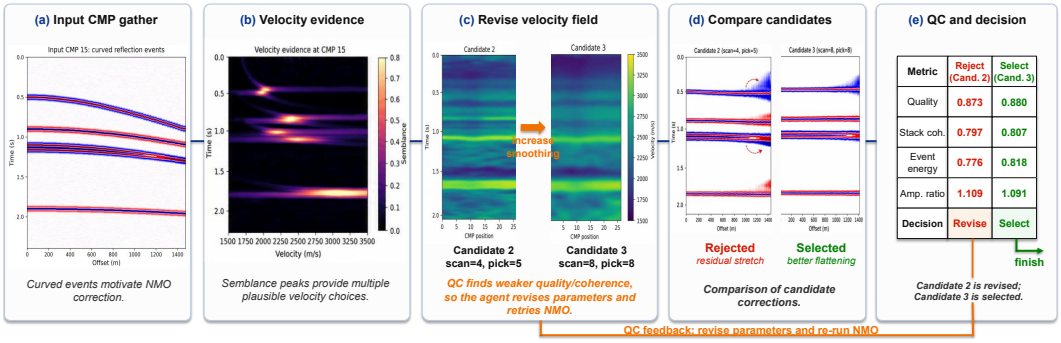


Figure 2: **Representative CMP normal-moveout (NMO) trajectory.** The agent uses semblance evidence to construct alternative velocity fields. Candidate 2 leaves residual stretch and weaker QC indicators; after increasing scan and pick smoothing, Candidate 3 produces flatter events and is selected. Orange and green arrows denote revision and termination, respectively.

corresponding reflection events become approximately horizontal across offset while event continuity and amplitude remain stable. Residual curvature, excessive stretch, or muting indicates that the velocity or correction parameters should be revised. The example illustrates uncertain parameter choice, intermediate evidence, revision, and stopping without requiring a detailed NMO derivation in the main text.

2.2 Agentic Seismic Processing

We model a task as

$$\tau = (g, \mathcal{X}_0, y_\tau^\star, \mathcal{F}_\tau, B_\tau, \mathcal{C}_\tau, \mathcal{E}_\tau), \tag{2}$$

where g is the natural-language goal, $\mathcal{X}_0$ is the set of named input artifacts, $y_\tau^\star$ is an optional hidden accepted target, $\mathcal{F}_\tau$ is the task-compatible operation space, B_τ is the interaction budget, $\mathcal{C}_\tau$ is the task contract, and $\mathcal{E}_\tau$ defines task-dependent evaluation. The accepted target, when present, is excluded from the policy observation.

At step t, the policy observes

$$o_t = (g, s(\mathcal{A}_t), h_t, \mathcal{F}_t, b_t)\,, \qquad \mathcal{F}_t \subseteq \mathcal{F}_\tau, \tag{3}$$

where $\mathcal{A}_t$ is the artifact workspace, $s(\mathcal{A}_t)$ contains compact scientific summaries, h_t is the execution history, and $b_t = B_\tau - t$ is the remaining budget. The action space is

$$a_t \in \{(f, \boldsymbol{\phi}) : f \in \mathcal{F}_t\} \cup \{\mathrm{finish}(\hat{x}) : \hat{x} \in \mathcal{A}_t\}\,. \tag{4}$$

The arguments $\boldsymbol{\phi}$ include input and output artifact identifiers as well as numerical and structured processing parameters. A successful processing action adds a new artifact and its lineage record to $\mathcal{A}_t$; a finish action selects an existing artifact as the output. A completed task therefore yields a processing trajectory ξ and a selected artifact $\hat{x}$.

3 SeisAgentBench

SeisAgentBench treats $(\xi, \hat{x})$, rather than a terminal string or file path alone, as its evaluation object. It separately checks whether the trajectory is admissible, whether the selected artifact satisfies the scientific objective, whether useful signal and structure are preserved, and whether the workflow uses its interaction budget reliably. Legal execution alone is therefore insufficient.

3.1 Benchmark Composition

SeisAgentBench contains 95 executable workflow tasks across 19 seismic-processing families, with five tasks per family. A task is not an isolated prediction point: it requires numerical processing over seismic artifacts, with decisions about operations, parameters, artifact dependencies, and final output selection. Parameter-sensitive families additionally require candidate assessment and revision. The suite is organized to cover distinct scientific objectives and failure modes across gathers, spectra, geometry products, stacked sections, physical models, and migrated images. *The benchmark's value lies in 95 executable assessments of scientific processing decisions, rather than merely 95 labeled samples.* Seismic arrays retain physical axes, sampling intervals, units, and semantic roles in regularly sampled format (RSF); geometry tasks may additionally use SEG-Y inputs. Figure 1 and Table 1 summarize the coverage.

Table 1: Task taxonomy of SeisAgentBench. Each family contains five tasks. Nine families use iterative candidate–assess–revise protocols, while ten use shorter executable artifact-construction protocols.

Scientific group	Families	Main products	Protocol composition
Signal conditioning	Automatic gain control; coherence enhancement; first-break and early-noise muting; F–K filtering; ground-roll suppression; spiking deconvolution; Radon-domain demultiple	Conditioned gathers and sections	6 iterative, 1 construction
Velocity and moveout	Spatially varying NMO correction; CMP NMO correction; NMO and CMP stacking; stacking-velocity analysis	Velocity fields, corrected gathers, and stacks	3 iterative, 1 construction
Geometry and statics	Datum-static correction; SEG-Y geometry binning; shot-to-CMP sorting	Corrected gathers and regularized geometry products	3 construction
Analysis and imaging	Frequency–wavenumber spectrum analysis; prestack time migration; zero-offset reverse-time migration	Frequency–wavenumber spectra and migrated images	3 construction
Physical models	Velocity-model construction; density-model construction	Velocity and density models	2 construction

The benchmark uses two task protocols. An *iterative protocol* requires candidate generation, candidate-specific assessment, revision after observed feedback, and supported final selection. An *artifact-construction protocol* requires an executable scientific sequence and a valid requested artifact, but does not impose repeated candidate search. The iterative requirement is applied only where parameter uncertainty is a central part of the scientific task. For example, filtering, gain control, moveout correction, and velocity analysis may admit several plausible parameter settings whose effects must be compared. Geometry conversion or spectrum construction, by contrast, may have a more direct scientific sequence. This division avoids making every episode artificially long while retaining explicit revision in the families where intermediate evidence is most informative.

The benchmark does not provide a canonical processing sequence. Different operation orders and parameter values may satisfy the same objective, provided that the final artifact and its lineage meet the task requirements. This design reflects the non-uniqueness of seismic processing: several outputs may be acceptable even when they do not exactly match one scripted workflow or one reference artifact.

The benchmark's base velocity and wavefield inputs include products derived from SubsurfaceGen (Stitt et al., 2026), which generates field-scale, geologically realistic synthetic Earth models and seismic responses. The use of structured Earth models and wave-equation simulation follows **established industrial practice** for validating seismic processing and imaging, as exemplified by SEAM Phase I (Fehler & Keliher, 2011). SeisAgentBench adds the processing-task layer: scientific objectives, typed operations, artifact dependencies, accepted outputs, and preservation criteria. This connects physically grounded seismic inputs to reproducible assessments of how an agent constructs and evaluates processing results. Appendix A describes task construction, and Appendix B documents the family-specific decisions and scientific criteria.

Seismic-processing researchers **designed and reviewed** the task objectives, accepted targets, parameter ranges, and evaluation criteria according to established domain practice. Positive qualitative feedback from clients and practitioners at major operators further supports the framework's practical relevance and consistency with expert processing practice.

3.2 Executable Processing Environment

At each step, the policy emits either one schema-constrained processing operation with numerical or structured parameters, or a finish action that selects an existing artifact. The global registry contains 88 typed interface names, including aliases and alternative backends rather than 88 independent seismic algorithms. Each task exposes only a scientifically relevant subset, and state-dependent preconditions enable an interface only when its required inputs exist. This structure prevents irrelevant or invalid calls without prescribing the operation sequence, numerical parameters, artifact dependencies, revision path, or stopping decision. Registry counts, representative interfaces, and backend distinctions are reported in Appendix C.5 (Tables 8 and 9).

The environment maintains a persistent artifact workspace rather than a single mutable array. Each generated artifact records its semantic role, parent artifacts, producing operation, parameters, and compact scientific summary. The resulting lineage graph allows an agent to compare candidates, branch from an earlier artifact, and return to a previous state. It also allows the evaluator to verify that a selected output was derived from the task inputs through a meaningful processing sequence.

Operation results expose compact evidence instead of serializing full multidimensional arrays into the language context. Depending on the artifact and operation, this evidence includes shape and physical axes, amplitude and energy statistics, spectral summaries, operation status, and candidate-specific scientific diagnostics. The next observation combines these summaries with recent actions, available candidates, finish blockers, and the remaining interaction budget. The interaction therefore requires four linked decisions: which operation to apply, how to parameterize it, whether to revise the result, and when to terminate. A concrete policy-visible prompt and evidence-driven revision example are shown in Appendix C.3 (Figure 4).

A finish action is accepted only when the selected artifact satisfies the task contract. Basic checks require readable and finite content, compatible shape and metadata, the requested artifact role, and valid input-derived lineage. They reject trivial or malformed solutions such as unchanged input copies, blank outputs, invalid geometry, or analysis-only products returned for a transformation task. For iterative families, the contract additionally requires materially different candidates, explicit candidate assessment, revision after observed feedback, and selection of an evaluated artifact. The action format and validation rules are given in Appendix C.2, while Appendix D specifies the basic artifact checks and minimum requirements for candidate construction, assessment, and revision.

3.3 Evaluation Protocol

SeisAgentBench evaluates each completed task using deterministic, family-specific seismic criteria grounded in established processing objectives and known physical failure modes. For task τ, selected artifact $\hat{x}$, and trajectory ξ, the benchmark reports

$$\mathbf{M}_\tau(\hat{x}, \xi) = (V_\tau, \mathbf{d}_\tau, \mathbf{p}_\tau, \mathbf{c}_\tau)\,, \tag{5}$$

where V_τ denotes valid completion, $\mathbf{d}_\tau$ contains the scientific measurements available for the requested product, $\mathbf{p}_\tau$ records artifact and preservation failures, and $\mathbf{c}_\tau$ records workflow behavior and cost.

The scientific vector $\mathbf{d}_\tau$ includes family-specific diagnostics and, when an accepted target $y_\tau^\star$ is available, reference-agreement measurements. The family-specific components retain the physical meaning of the requested processing operation: NMO measures event flatness, coherence, offset support, amplitude preservation, and stretch; filtering measures target-noise attenuation together with reflection or pass-band preservation; deconvolution measures compactness, sidelobes, spectral leakage, and energy preservation; geometry tasks measure trace conservation and fold; and model or imaging tasks measure physical range, structural consistency, focus, and positioning. Reference-agreement components include correlation, normalized root-mean-square error (NRMSE), structural similarity index measure (SSIM), peak signal-to-noise ratio (PSNR), amplitude agreement, input-relative improvements, and shape validity when applicable (Wang et al., 2004).

For a compact measure of joint workflow completion and artifact agreement, the evaluator summarizes the fixed reference-agreement components recorded in $\mathbf{d}_\tau$ and applies the task-completion indicator:

$$q_\tau = V_\tau\, S_\tau(\mathbf{d}_\tau)\,, \tag{6}$$

where S_τ is fixed before test evaluation and invalid tasks receive zero. Not every family-specific diagnostic must enter this scalar: measurements that are meaningful only within one family remain available for per-family reporting and scientific interpretation. The reference-agreement components, their weights, missing-component handling, and failure-aware aggregation used in the experiments are specified in Appendix E.2.

Let $\mathcal{F}$ denote the 19 families, let $\mathcal{F}_{\text{iter}} \subset \mathcal{F}$ denote the nine iterative families, and let $\mathcal{T}_f$ contain the evaluated tasks from family f. We define

$$Q_{\text{sci}} = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \frac{1}{|\mathcal{T}_f|} \sum_{\tau \in \mathcal{T}_f} q_\tau, \tag{7}$$

and

$$Q_{\text{iter}} = \frac{1}{|\mathcal{F}_{\text{iter}}|} \sum_{f \in \mathcal{F}_{\text{iter}}} \frac{1}{|\mathcal{T}_f|} \sum_{\tau \in \mathcal{T}_f} q_\tau. \tag{8}$$

Table 2: Task-level evaluation in SeisAgentBench. Detailed measurements retain the scientific meaning of each product, while q_τ provides the common task score summarized by Q_{sci} and Q_{iter}.

Component	Scientific meaning and reported use
V_τ	Successful finish; requested output role; readable and finite content; shape and metadata consistency; input-derived lineage; required workflow conditions.
$\mathbf{d}_\tau$	Family-specific scientific diagnostics and compatible reference-agreement measurements. Examples include NMO flatness and stretch, filtering attenuation and preservation, deconvolution compactness and spectral stability, geometry and fold, model plausibility, migration focus, correlation, NRMSE, SSIM, PSNR, amplitude agreement, and shape validity.
$\mathbf{p}_\tau$	Blank or collapsed output; over-muting; gain instability; amplitude or event-energy loss; residual coherent noise; ringing; edge artifacts; shape or geometry damage.
$\mathbf{c}_\tau$	Total and meaningful operations; generated candidates; explicit evaluations; post-QC revisions; invalid actions; blocked finishes; execution failures; budget use.
q_τ	Failure-aware scalar summary of the normalized automatic components selected from $\mathbf{d}_\tau$. Invalid tasks receive zero. Q_{sci} and Q_{iter} aggregate this score over all families and the iterative families, respectively.

Thus, q_τ summarizes the automatic quality of one completed task, Q_{sci} summarizes the same task score across all 19 families, and Q_{iter} summarizes it across the nine iterative families. On the held-out test split, each family contributes one task.

The scientific objective and preservation checks capture complementary requirements: NMO should flatten reflections without excessive stretch or amplitude loss, while filtering should attenuate unwanted energy without removing useful reflections. Recording these effects separately exposes destructive processing that successful execution or a single agreement score could overlook. Artifact-warning rate summarizes invalid tasks and prespecified preservation failures, derived from V_τ and $\mathbf{p}_\tau$, while meaningful operations and post-QC revisions, both derived from $\mathbf{c}_\tau$, describe the processing trajectory. Appendices E.4–E.7 provide the definitions, thresholds, and task-aware visualizations.

4 SeisAgent

SeisAgent is a causal language-model policy $\pi_\theta(a_t \mid o_t)$ specialized for the executable interaction protocol defined above. The language model directly selects processing actions, while the environment maintains long-horizon state through the artifact workspace, lineage, interaction history, and candidate records. Training separates two problems: learning how to interact correctly with the environment and learning which valid processing decisions produce stronger seismic artifacts.

4.1 Trajectory-Level Supervised Warm Start

We initialize $\pi_\theta(a_t \mid o_t)$ using a small trajectory-level SFT warm start. Executable workflows are run in the same benchmark environment, retained when they satisfy the verifier, and converted into turn-level observation–action pairs. The policy is trained by minimizing the negative log likelihood of the recorded action,

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(o,a^\star)\sim\mathcal{D}_{\mathrm{SFT}}}\left[\log \pi_\theta(a^\star \mid o)\right]. \tag{9}$$

The warm start teaches schema-constrained action generation, artifact references, interpretation of compact data summaries, basic candidate inspection, and valid termination. It is deliberately small: it does not provide workflows for the complete benchmark and is not presented as a set of expert demonstrations. Its purpose is to establish interaction competence so that subsequent rollouts contain enough valid scientific behavior to receive informative feedback. The supervised task counts and training comparisons are reported in Sec. 5.1 and Table 3.

4.2 Scientific-Feedback Group-Relative Post-Training

After the supervised warm start, SeisAgent is post-trained through interaction with the executable environment. The step reward separates execution validity, scientific improvement, artifact preservation, and interaction cost:

$$r_t = r_t^{\mathrm{exec}} + r_{\tau,t}^{\mathrm{quality}} - \lambda_p p_{\tau,t}^{\mathrm{art}} - \lambda_c c_t. \tag{10}$$

The execution term r_t^{exec} rewards parseable, state-valid operations, successful execution, and supported completion. It penalizes malformed or unavailable operations, execution failures, unsupported finish attempts, missing outputs, and terminal failure. A finish is supported when the selected artifact satisfies the task contract, including its required role, lineage, validity checks, and, for iterative families, candidate-assessment requirements.

For the candidate produced at step t, the evaluator computes the available scientific measurements $\mathbf{d}_{\tau,t}$. The quality term $r_{\tau,t}^{\mathrm{quality}}$ combines improvements in reference-free candidate diagnostics and private reference agreement, together with the configured first-measurement and new-best bonuses;

its component weights are included in the term. For example, NMO feedback measures event flatness, coherence, offset support, and stretch control, while filtering feedback measures target-noise attenuation together with reflection preservation. These measurements provide feedback during processing, before the terminal completion requirements are satisfied. At termination, the selected artifact is scored by q_τ and summarized by Q_{sci} and Q_{iter}. Training and reporting therefore use the same scientific measurement infrastructure, with their respective weighting and completion rules specified in Appendices F.1 and E.2.

The artifact term $p^{\mathrm{art}}_{\tau,t}$ is a nonnegative scalar derived from the preservation checks for the current candidate:

$$p^{\mathrm{art}}_{\tau,t} = G_\tau(\mathbf{p}_{\tau,t}), \tag{11}$$

where $\mathbf{p}_{\tau,t}$ is the candidate-level counterpart of $\mathbf{p}_\tau$ in Eq. 5, and G_τ aggregates the applicable violations using fixed task weights. Missing outputs, malformed actions, and failed execution are handled by r^{exec}_t; $p^{\mathrm{art}}_{\tau,t}$ applies to produced candidates that pass basic execution checks but violate signal- or structure-preservation criteria. Depending on the family, these violations include near-blank or collapsed artifacts that remain file-valid, over-muting, unstable gain, amplitude or event-energy loss, residual coherent noise, ringing, edge artifacts, and shape or geometry damage. Appendix F.1 reports the implemented penalty terms and their coefficients.

The scientific-improvement and artifact terms are kept separate because improving the target criterion does not guarantee signal preservation. For example, an NMO candidate may improve event flatness while introducing stretch or amplitude loss, and a filtering candidate may suppress target noise while removing useful reflections. The reward balances improvement in the requested processing effect against collateral damage to useful signal and structure.

Finally, c_t measures the interaction cost of the current decision. It includes the cost of consuming a step or operation call, with additional cost for late or redundant interaction. Invalid and failed operations are handled primarily by r^{exec}_t, while c_t discourages unnecessary processing after a sufficiently supported candidate has been found. The implementation also provides a bounded terminal benefit for unused budget; its coefficient and the other reward coefficients are listed in Table 11.

The reward form is shared across all families. Task dependence enters through the scientific measurements $\mathbf{d}_{\tau,t}$, the task scoring rule, and the preservation checks $\mathbf{p}_{\tau,t}$, rather than through a separate RL algorithm for each family. During training, hidden reference agreement may contribute privately to the scientific and preservation terms when an accepted target is available, but neither the target nor any reference-derived value enters the policy observation. For each task, we sample four complete trajectories and standardize their discounted returns within the same-task group. The policy is updated with a clipped old-policy likelihood-ratio objective using group-relative advantages and transition-level process credit, without a learned value model. Appendix F.2 specifies the complete objective and optimization settings.

5 Experiments

Following the benchmark design, our experiments evaluate three properties that completion-only benchmarks would mix together: whether a policy returns a valid seismic artifact, whether the artifact is scientifically consistent with the requested processing objective, and whether the policy uses intermediate evidence to construct, revise, and terminate a workflow. We first study model scale under one fixed post-training recipe, then compare trajectory supervision and Group-RL configurations, and finally examine how learning changes workflow depth and operation-specific seismic diagnostics.

5.1 Experimental Setup

Benchmark and split. We use the family-balanced 57/19/19 train, development, and test split of the 95 SeisAgentBench tasks. Checkpoints are selected on the development split and evaluated once on the held-out test split using deterministic decoding. This split tests held-out tasks from all 19 known families; the three training seeds measure optimization variability rather than variation across independent surveys. Some underlying artifacts are shared across partitions, so the evaluation is not source-disjoint. During evaluation, the policy observes the task instruction, available operation schemas, artifact summaries, interaction history, candidate diagnostics, and remaining budget, but not the accepted target or target-derived intermediate scores. The policy information boundary and rendered-prompt audit are documented in Appendix A.3.

Table 3: Results on the 19-task held-out test split, with one task from each seismic-processing family. Block A compares backbones under zero-shot interaction, schema ICL-2, and one SFT-2+Group-RL recipe selected on Qwen3-14B development data and then fixed across backbones. Block B compares training configurations on Qwen3-14B, and Block C reports deterministic controls. Trainable conditions show mean ± sample standard deviation over three seeds.

		Training exposure		Automatic held-out outcomes					Workflow behavior	
Backbone	Method	SFT tasks	RL tasks / traj.	Valid (%)↑	Q_{sci} ↑	Q_{iter} ↑	ΔSSIM (n)↑	Artifact warn. (%)↓	Meaningful ops	Post-QC rev.
A. Backbone comparison under one frozen SFT-2 + Group-RL recipe										
Qwen3-4B	Zero-shot	0	0	78.9	0.551	0.421	0.222 (8)	42.1	4.79	1.16
	Schema ICL-2	0	0	84.2	0.624	0.462	0.237 (8)	47.4	5.26	1.11
	SFT-2 + Group-RL	2	55 / 220	84.2 ± 5.3	0.675 ± 0.037	0.460 ± 0.078	0.244 ± 0.015 (8)	40.4 ± 3.0	5.00 ± 0.27	1.14 ± 0.15
Qwen3-8B	Zero-shot	0	0	84.2	0.742	0.507	0.271 (8)	36.8	5.68	1.47
	Schema ICL-2	0	0	89.5	0.680	0.472	0.263 (9)	36.8	6.79	1.63
	SFT-2 + Group-RL	2	55 / 220	89.5 ± 0.0	0.733 ± 0.009	0.586 ± 0.020	0.279 ± 0.015 (9)	31.6 ± 0.0	6.32 ± 0.14	1.56 ± 0.06
Qwen3-14B	Zero-shot	0	0	94.7	0.745	0.775	0.213 (10)	21.1	5.26	1.26
	Schema ICL-2	0	0	100.0	0.744	0.754	0.215 (11)	26.3	5.63	1.26
	SFT-2 + Group-RL	2	55 / 220	100.0 ± 0.0	0.750 ± 0.004	0.765 ± 0.008	0.221 ± 0.002 (11)	21.1 ± 0.0	5.30 ± 0.08	1.26 ± 0.00
Qwen3-32B	Zero-shot	0	0	94.7	0.747	0.812	0.218 (10)	26.3	5.89	1.32
	Schema ICL-2	0	0	94.7	0.752	0.836	0.217 (10)	26.3	6.00	1.21
	SFT-2 + Group-RL	2	55 / 220	100.0 ±0.0	0.767 ±0.017	0.855 ±0.035	0.239 ±0.012(11)	24.6 ±3.0	5.81 ±0.08	1.26 ±0.05
Llama-3.1-8B-Instruct	Zero-shot	0	0	63.2	0.512	0.331	0.114 (5)	52.6	4.74	1.37
	Schema ICL-2	0	0	73.7	0.573	0.369	0.167 (8)	47.4	5.21	1.16
	SFT-2 + Group-RL	2	55 / 220	84.2 ± 5.3	0.676 ± 0.044	0.461 ± 0.067	0.189 ± 0.029 (8)	42.1 ± 5.3	4.44 ± 0.16	1.21 ± 0.42
Mistral-7B-Instruct-v0.3	Zero-shot	0	0	52.6	0.420	0.195	0.220 (5)	63.2	6.89	1.58
	Schema ICL-2	0	0	68.4	0.502	0.204	0.121 (5)	63.2	7.26	1.53
	SFT-2 + Group-RL	2	55 / 220	75.4 ± 3.0	0.584 ± 0.019	0.443 ± 0.115	0.184 ± 0.060 (8)	47.4 ± 5.3	5.18 ± 0.41	1.05 ± 0.14
B. Training-configuration comparison on Qwen3-14B										
Qwen3-14B	SFT-1 only	1	0	94.7 ± 0.0	0.740 ± 0.003	0.766 ± 0.005	0.222 ± 0.001 (10)	21.1 ± 0.0	5.44 ± 0.08	1.26 ± 0.00
Qwen3-14B	SFT-2 only	2	0	98.2 ± 3.0	0.741 ± 0.022	0.746 ± 0.046	0.217 ± 0.004 (11)	21.1 ± 5.3	5.65 ± 0.56	1.28 ± 0.03
Qwen3-14B	Group-RL only	0	55 / 220	100.0 ± 0.0	0.745 ± 0.002	0.765 ± 0.003	0.219 ± 0.000 (11)	21.1 ± 0.0	5.25 ± 0.13	1.26 ± 0.00
Qwen3-14B	SFT-1 + Group-RL	1	55 / 220	98.2 ± 3.0	0.747 ± 0.019	0.748 ± 0.040	0.220 ± 0.011 (11)	21.1 ± 0.0	5.42 ± 0.11	1.33 ± 0.08
Qwen3-14B	SFT-2 + Group-RL	2	55 / 220	100.0 ± 0.0	0.750 ± 0.004	0.765 ± 0.008	0.221 ± 0.002 (11)	21.1 ± 0.0	5.30 ± 0.08	1.26 ± 0.00
C. Fixed-script execution controls										
Scripted	Family-default workflow	0	0	47.4	0.406	0.000	0.082 (2)	63.2	3.84	0.00
Scripted	Reference-visible workflow	0	0	47.4	0.406	0.000	0.082 (2)	63.2	4.68	0.00

Notes. SFT tasks count unique supervised tasks; RL exposure reports unique training tasks and complete trajectories. Q_{sci} averages the failure-aware task score q_τ across all 19 families, while Q_{iter} applies the same aggregation to the nine iterative families. Invalid tasks receive zero. ΔSSIM is computed for 11 semantically compatible families, and each cell reports the valid denominator n. Artifact warn. counts invalid tasks and valid tasks with at least one prespecified artifact or preservation failure. Meaningful operations and post-QC revisions are descriptive. The scripts are fixed execution controls without candidate revision; they therefore do not satisfy the iterative completion contract. The reference-visible script is a sensitivity control, not an adaptive-search baseline or performance ceiling.

Models and training. We compare Qwen3 backbones (Yang et al., 2025) at 4B, 8B, 14B, and 32B scales and include Llama-3.1-8B-Instruct (Grattafiori et al., 2024) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) as cross-model-family portability controls, under zero-shot interaction, two-example schema in-context learning (ICL-2) (Brown et al., 2020), and one SFT-2+Group-RL recipe selected using Qwen3-14B development data and then fixed across scales. On Qwen3-14B, we further compare SFT-1, SFT-2, Group-RL only, SFT-1+Group-RL, and SFT-2+Group-RL. SFT uses one or two unique training tasks as a small interaction warm start. All Group-RL conditions use the same 55-task pool, with five stages of 11 task groups and four trajectories per group, yielding 220 complete rollouts in total. Every trainable condition is evaluated over three seeds. Group-RL optimization settings are given in Appendix F.2. Appendix G reports the experimental matrix, LoRA settings (Hu et al., 2022), and hardware and software configurations.

Metrics. Each test task produces the evaluation record in Eq. 5. Valid reports the mean of V_τ. Q_{sci} and Q_{iter} are the family-level summaries of the final task score q_τ defined in Eqs. 7 and 8. We additionally report input-relative ΔSSIM for the 11 families whose input and output products permit a meaningful aligned comparison. Artifact-warning rate counts invalid tasks and valid tasks with at least one prespecified failure in $\mathbf{p}_\tau$. Meaningful operations and post-QC revisions are components of $\mathbf{c}_\tau$ and describe workflow behavior without an unconditional better direction. Reference-agreement and workflow measures are detailed in Appendices E.2–E.4; Table 10 summarizes the scientific evidence used for each family.

5.2 Held-Out Results

Table 3 shows three main findings.

Model scale strengthens the starting policy, but does not remove the scientific-quality challenge. Zero-shot validity rises from 78.9% for Qwen3-4B to 94.7% for Qwen3-14B and Qwen3-32B, while Q_{sci} increases from 0.551 to approximately 0.75. The artifact-level measures, however, continue to distinguish both model scales and training methods after validity becomes high. For example, the trained Qwen3-32B policy reaches 100% validity with $Q_{\mathrm{sci}} = 0.767 \pm 0.017$ and $Q_{\mathrm{iter}} = 0.855 \pm 0.035$, compared with 94.7%, 0.747, and 0.812 for its zero-shot counterpart. Thus,

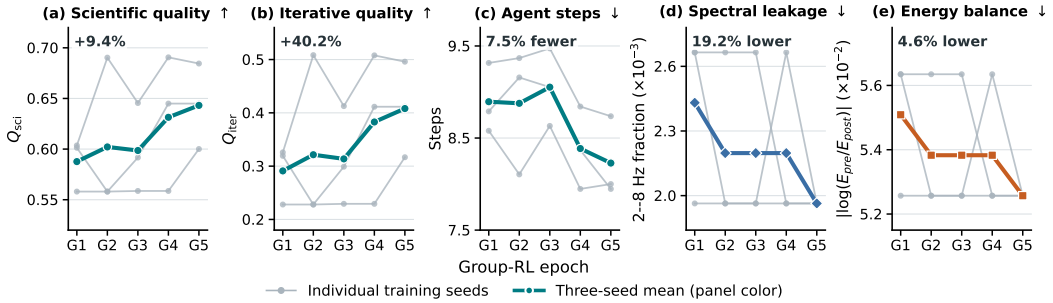


Figure 3: **Group-RL learning dynamics and seismic diagnostics.** Panels (a–c) report development-set automatic quality, iterative-family quality, and environment steps across five Group-RL epochs for Qwen3-4B. Panels (d–e) report low-frequency spectral leakage and energy-ratio error for a representative spiking-deconvolution task.

larger models are more reliable at constructing an admissible product, but post-training still changes the scientific agreement and iterative-family quality of the returned artifact.

Post-training changes workflow depth according to model capacity. For the smaller models, improved processing is associated with spending more actions on constructing and checking candidate artifacts. At 4B, SFT-2+Group-RL raises Q_{sci} from 0.551 to 0.675 ± 0.037 while increasing meaningful operations slightly from 4.79 to 5.00 ± 0.27. At 8B, the trained policy improves Q_{iter} from 0.507 to 0.586 ± 0.020 and reduces the artifact-warning rate from 36.8% to 31.6%, while meaningful operations increase from 5.68 to 6.32 ± 0.14. These results suggest that smaller zero-shot policies often stop after an insufficiently assessed workflow, and that post-training teaches them to allocate additional actions to candidate construction, QC, and parameter revision. The direction changes for larger models. At 14B, SFT-2+Group-RL obtains $Q_{\mathrm{sci}} = 0.750 \pm 0.004$ with 5.30 ± 0.08 meaningful operations, compared with 0.744 and 5.63 operations for ICL-2. At 32B, the trained policy improves Q_{sci} from 0.752 to 0.767 ± 0.017 relative to ICL-2 while reducing meaningful operations from 6.00 to 5.81 ± 0.08. These patterns are consistent with the reward design: additional processing can improve insufficiently developed candidates, while efficient candidate selection can reduce unnecessary calls once a strong workflow is available.

Group-RL supports effective workflow specialization. The SFT warm start initializes the interaction protocol, while Group-RL trains the policy to prefer higher-return executable workflows. Group-RL conditions obtain stronger validity or automatic artifact quality than SFT-only training while using fewer meaningful operations than SFT-2 alone. Adding the two-task SFT warm start gives a small further improvement in the all-family score, but Group-RL alone is already competitive on the iterative families. Schema ICL can improve action-format compliance, but its effect on scientific quality is inconsistent across model sizes. These results support evaluating the produced artifact and workflow behavior rather than judging an agent only by successful termination.

5.3 Learning Dynamics and Scientific Diagnostics

Figure 3 shows that, from the first to the fifth Group-RL epoch, the three-seed mean Q_{sci} for Qwen3-4B increases by 9.4%, Q_{iter} increases by 40.2%, and the mean number of environment steps decreases by 7.5%. This joint trend is consistent with increasingly effective workflows that improve the benchmark scores while using fewer environment steps. The deconvolution case provides a domain-level check of this interpretation: the 2–8 Hz spectral-leakage fraction decreases by 19.2%, while the absolute log energy-ratio error decreases by 4.6%. The former indicates less undesired low-frequency energy after deconvolution, and the latter indicates better preservation of total signal energy. The improvement is therefore visible not only in the aggregate score, but also in physical properties associated with the requested seismic operation. Representative outputs across four task families are shown in Appendix E.7 (Figure 5). A complete CMP NMO trajectory recorded during development is provided in Appendix H (Table 15).

6 Conclusion and Limitations

SeisAgentBench and SeisAgent establish executable evaluation and scientific-feedback learning across 19 seismic-processing families. By assessing processing trajectories together with their resulting artifacts, the framework makes numerical decisions, preservation trade-offs, and workflow revision explicit targets for agent research. The current evaluation covers held-out tasks within known families; independent field-survey transfer is a direction for further validation.

AI USE STATEMENT

LLMs are used to assist with grammar checking, improving writing fluency, managing computational resources and providing suggestions after the initial draft. All observations, ideas, methodologies in this paper are developed entirely by the authors. Any content generated with the help of LLMs is created under detailed author instructions and thoroughly verified by the authors before inclusion.

REPRODUCIBILITY STATEMENT

Benchmark construction, task definitions, operation interfaces, and completion contracts are documented in Appendices A–D. Appendix E describes the evaluation metrics and scientific diagnostics, while Appendices F and G report reward coefficients, optimization settings, model configurations, and hardware and software environments. Trainable conditions are evaluated over three seeds, with development-based checkpoint selection and deterministic held-out evaluation. Representative prompts, processing outputs, and a complete trajectory illustrate the executable protocol. Upon acceptance, we will publicly release the full SeisAgentBench benchmark, including task data and definitions, data splits, executable processing interfaces, and evaluation code, together with the SeisAgent training and evaluation scripts.

REFERENCES

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

Kuangshi Ai, Haichao Miao, Kaiyuan Tang, Nathaniel Gorski, Jianxin Sun, Guoxi Liu, Helgi I Ingolfsson, David Lenz, Hanqi Guo, Hongfeng Yu, et al. Scivisagentbench: A benchmark for evaluating scientific data analysis and visualization agents. *arXiv preprint arXiv:2603.29139*, 2026.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, Botao Yu, Yifei Li, Zeyi Liao, Chen Wei, Zitong Lu, et al. Scienceagentbench: Toward rigorous assessment of language agents for data-driven scientific discovery. In *International Conference on Learning Representations*, volume 2025, pp. 96934–96990, 2025.

C Hewitt Dix. Seismic velocities from surface measurements. *Geophysics*, 20(1):68–86, 1955.

Michael Fehler and P Joseph Keliher. *SEAM phase 1: Challenges of subsalt imaging in tertiary basins, with emphasis on deepwater Gulf of Mexico*. Society of Exploration Geophysicists, 2011.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Edward J Hu, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.

Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations*, volume 2024, pp. 54107–54157, 2024.

Rayan Kanfar, Abdulmohsen Alali, Thierry-Laurent Tonellot, Hussain Salim, and Oleg Ovcharenko. Intelligent seismic workflows: The power of generative ai and language models. *The Leading Edge*, 44(2):142–151, 2025.

Farbod Khosro Anjom, Francesco Vaccarino, and Laura Valentina Socco. Machine learning for seismic exploration: Where are we and how far are we from the holy grail? *Geophysics*, 89(1): WA157–WA178, 2024.

Jiarui Lu, Thomas Holleis, Yizhe Zhang, Bernhard Aumayer, Feng Nan, Haoping Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, et al. Toolsandbox: A stateful, conversational, interactive evaluation benchmark for llm tool use capabilities. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 1160–1183, 2025.

W Harry Mayne. Common reflection point horizontal data stacking techniques. *Geophysics*, 27(6): 927–938, 1962.

Norman S Neidell and M Turhan Taner. Semblance and other coherency measures for multichannel data. *Geophysics*, 36(3):482–497, 1971.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35: 27730–27744, 2022.

Enders A Robinson. Predictive decomposition of seismic traces. *Geophysics*, 22(4):767–778, 1957.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems*, 36:68539–68551, 2023.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Joseph Stitt, Pratik Rathore, Madeleine Udell, and Ching-Yao Lai. Subsurfacegen: Procedural generation of field-scale earth models and seismic data. *arXiv preprint arXiv:2605.30541*, 2026.

Weiwei Sun, Shengyu Feng, Shanda Li, and Yiming Yang. Co-bench: Benchmarking language model agents in algorithm search for combinatorial optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 33126–33134, 2026.

M Turhan Taner and Fulton Koehler. Velocity spectra—digital computer derivation applications of velocity functions. *Geophysics*, 34(6):859–881, 1969.

Jiangyuan Wang, Kejun Xiao, Qi Sun, Huaipeng Zhao, Tao Luo, Jian Dong Zhang, and Xiaoyi Zeng. Shoppingbench: A real-world intent-grounded shopping benchmark for llm-based agents. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 33521–33529, 2026.

Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004.

Matthew J Widlansky and Nemanja Komar. Building an intelligent data exploring assistant for geoscientists. *Journal of Geophysical Research: Machine Learning and Computation*, 2(3): e2025JH000649, 2025.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. Webshop: Towards scalable real-world web interaction with grounded language agents. *Advances in Neural Information Processing Systems*, 35:20744–20757, 2022a.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *NeurIPS 2022 Foundation Models for Decision Making Workshop*, 2022b.

Öz Yilmaz. *Seismic data analysis: Processing, inversion, and interpretation of seismic data*. Society of exploration geophysicists, 2001.

Siwei Yu and Jianwei Ma. Deep learning for geophysics: Current and future trends. *Reviews of Geophysics*, 59(3):e2021RG000742, 2021.

A Benchmark Construction and Release Details

A.1 Benchmark identity and scope

SeisAgentBench contains 95 executable tasks from 19 seismic-processing families, with five tasks per family. Nine families use an iterative candidate–assess–revise protocol and ten use an artifact-construction protocol. The iterative subset therefore contains 45 tasks, while the artifact-construction subset contains 50 tasks. The task set manipulates typed seismic gathers, spectra, geometry products, stacks, velocity and density models, and migrated images. Most multidimensional arrays use regularly sampled format (RSF), which stores physical axis metadata in a text header and sample values in a binary sidecar. The geometry-binning family additionally uses SEG-Y-style trace records.

The suite is designed around controlled, physically structured seismic artifacts so that accepted targets, corruption processes, artifact checks, and family-specific diagnostics can be evaluated reproducibly. Task objectives, parameter ranges, accepted targets, and evaluator criteria follow established seismic-processing practice and were reviewed within the project by seismic-processing researchers. The benchmark is intended to test scientific workflow construction under controlled conditions rather than unattended field-survey production.

Table 4: Release-candidate benchmark and experimental scale.

Property	Value
Tasks	95
Families	19
Tasks per family	5
Iterative families / tasks	9 / 45
Artifact-construction families / tasks	10 / 50
Interaction budgets	6–28 calls
Primary split	57 train / 19 development / 19 test
Training seeds	2026, 2027, 2028
Typed public interfaces	88
Scientific processing/modeling groups	52

A.2 Task schema

Each task is represented as

$$\tau = (g, \mathcal{X}_0, y_\tau^\star, \mathcal{F}_\tau, B_\tau, \mathcal{C}_\tau, \mathcal{E}_\tau), \tag{12}$$

where g is the natural-language objective, $\mathcal{X}_0$ is the set of named input artifacts, $y_\tau^\star$ is an optional hidden accepted target, $\mathcal{F}_\tau$ is the task-compatible operation set, B_τ is the interaction budget, $\mathcal{C}_\tau$ is the task contract, and $\mathcal{E}_\tau$ specifies online and offline evaluation. The machine-readable task definition also records the requested output role, axis requirements, required operation groups, expected artifact lineage, and iterative-protocol fields when applicable.

Task construction follows three stages. First, input and accepted target products are assembled for each seismic family. Second, these products are compiled into goal-oriented task files with neutral workspace aliases; reference paths, task identifiers, prescribed operation sequences, and expected parameters are removed from the policy view. Third, each task is checked for readable arrays, finite values, compatible metadata, valid rendering, executable contracts, and reference isolation.

A.3 Primary split and policy information boundary

Development data are used for per-epoch checkpoint selection; a locked checkpoint is then evaluated once on the 19-task test split with deterministic decoding. This protocol measures held-out tasks from known families.

A rendered-prompt audit applies the actual prompt builder to all 95 tasks and checks instructions, artifact summaries, operation schemas, history, and finish blockers. The audited protocol reports 95/95 passes and no detected accepted-reference paths, hidden workflow targets, expected parameters, or reference-derived scores. The task-compatible operation subset can reveal the functional type of work; this is explicit environment structure rather than a hidden task label. A concrete policy-visible prompt and its evidence-driven revisions are shown in Figure 4.

Table 5: Information boundary. Private reference signals can influence training gradients but cannot enter the policy observation, test-time candidate ranking, finish legality, parameter selection, or stopping rule.

Policy-visible state	Private training verifier	Post-episode evaluator
Natural-language goal; neutral input and output aliases; shape, axes, units, amplitude, energy, and spectral summaries; state-valid operation schemas; bounded history; candidates; reference-free diagnostics; workflow counters; finish blockers; remaining budget	Hidden accepted artifact; reference SNR and normalized agreement; private artifact checks; scalar reward and reward breakdown; terminal verifier signals	Final correlation, normalized root-mean-square error (NRMSE), structural similarity index measure (SSIM), peak signal-to-noise ratio (PSNR), gains relative to input, residual and difference panels, task-aware quality-control summaries, and aggregate reports

B Complete Task-Family Catalog

Tables 6 and 7 list every family, its scientific purpose, the main decisions made by the policy, the scientific evidence used to assess outputs, common artifact risks, and the interaction budget. The listed decisions describe scientific choices rather than one required script; several operation orders and parameter values can satisfy the same task contract.

Table 6: The nine iterative families. Each family contains five tasks and requires candidate construction, candidate-specific assessment, revision after observed evidence, and selection of an evaluated candidate.

Family	Objective and product	Main decisions	Scientific evidence	Common artifact risks	Budget
Automatic gain control (agc_gain)	Balance amplitudes in a gather or section without amplifying quiet-zone noise	Gain window, scaling strength, and candidate selection	Robust-amplitude balance, adjacent-trace correlation, singular-value energy, active/quiet contrast, and quiet-zone fraction	Noise bloom, gain explosion, unstable amplitudes, and near-blank support	28
Coherence enhancement (coherence_enhancement)	Improve reflection continuity in a noisy gather or section	Enhancement window or strength and candidate choice	Flatness, stack coherence, adjacent-trace correlation, and trace-energy balance	Smearing, amplitude loss, and apparent coherence caused by collapse	12
Spatially varying normal moveout (NMO) correction (field_nmo_correction)	Apply time- and location-dependent moveout correction	Velocity scan range, temporal/spatial picks, smoothing, representation, and stretch control	Stack and adjacent coherence, amplitude, coverage, entropy, active support, event continuity, and collapse/stretch guards	Residual moveout, excessive stretch, mute loss, and event collapse	24
First-break and early-noise muting (first_break_mute)	Suppress direct or early arrivals while preserving reflection support	Mute boundary, taper, and candidate selection	Early-energy attenuation, live fraction, coherence, event support, and signal preservation	Over-muting, residual early noise, edge discontinuities, and loss of reflection support	12
Frequency–wavenumber filtering (fk_filtering)	Attenuate coherent low-apparent-velocity noise while preserving reflections	Velocity/frequency cutoffs, transition slope, and candidate selection	Low-apparent-velocity energy reduction, reflection or pass-band preservation, output stability, and ringing guards	Reflection loss, residual coherent noise, ringing, and amplitude instability	12
Ground-roll suppression (ground_roll_suppression)	Reduce dispersive ground roll while preserving the signal band	Low-velocity region, transition softness, and candidate selection	Low-velocity attenuation, signal-band preservation, transition stability, ringing, and amplitude guards	Residual ground roll, signal-band loss, ringing, and unstable scaling	12
Common-midpoint (CMP) NMO correction (nmo_correction)	Flatten primary reflection events across offset with controlled stretch	Constant or time-varying velocity, picks, smoothing, and stretch mute	Event flatness, stack coherence, offset support, amplitude preservation, and stretch/mute guards	Residual curvature, excessive stretch, amplitude loss, and over-mute	24
Spiking deconvolution (spiking_deconvolution)	Compress the source wavelet and improve temporal resolution without unstable ringing	Operator and prediction settings, stabilization, and candidate selection	Wavelet phase and width, compactness, low-frequency guard, sidelobes, sharpness, amplitude, crest factor, and coherence	Ringing, low-frequency leakage, energy distortion, amplitude loss, and unstable spikes	28
Stacking-velocity analysis (velocity_picking)	Infer a time-varying velocity field supported by CMP coherence evidence	Search range, velocity picks, temporal/spatial smoothing, and support control	Field-NMO quality, stack and adjacent coherence, coverage, entropy, active support, residual moveout, and pick smoothness	Noisy or discontinuous picks, over-smoothing, implausible velocities, and weak offset support	24

Table 7: The ten artifact-construction families. These tasks still require an executable scientific sequence and a valid requested product, but they do not force repeated candidate search when the scientific path is more direct.

Family	Objective and product	Main decisions	Scientific evidence	Common artifact risks	Budget
Datum-static correction (datum_statics)	Correct datum shifts and improve event alignment	Shift estimates, application order, and output role	Event alignment, static-shift profile, shape, metadata, and reference agreement	Residual misalignment, edge effects, wrong shift sign, and metadata damage	8
Density-model construction (density_model_building)	Derive a physically plausible density model from velocity information	Velocity-density relation, smoothing, and admissible range	Density range, velocity-density relation residual, structural consistency, correlation, NRMSE, and SSIM	Nonphysical values, discontinuities, over-smoothing, and structural mismatch	8
Frequency–wavenumber spectrum analysis (fk_spectrum_analysis)	Construct a valid F–K spectrum with correct axes and energy distribution	Analysis axes, transform settings, and output normalization	Shape and axis validity, spectral-energy localization, and deterministic target agreement	Swapped axes, invalid normalization, wrong shape, and misleading spectral support	6
SEG-Y geometry binning (geometry_binning)	Map irregular traces and headers to a regular spatial product	Header fields, bin definitions, spatial mapping, and output geometry	Trace and sample conservation, fold range, filled-bin fraction, ordering, axes, and metadata validity	Lost or duplicated traces, empty bins, invalid fold, and incorrect geometry	8
NMO correction and CMP stacking (nmo_stack)	Correct moveout and combine offset traces into a stacked product	Velocity input, correction, stacking operation, and artifact dependencies	Coherence, event continuity, retained variance, shape, and reference agreement	Smearing, destructive stacking, amplitude cancellation, and wrong dimensions	14
Prestack time migration (prestack_migration)	Focus prestack reflections using a velocity model	Velocity/model input, imaging parameters, aperture, and processing order	Image focus, edge artifacts, positioning, amplitude stability, shape, and reference agreement	Defocus, edge energy, positioning error, and invalid geometry	10
Radon-domain demultiple (radon_demultiple)	Attenuate multiples while preserving primary reflections	Transform parameters, regularization, component selection, and reconstruction	Multiple attenuation, primary preservation, component balance, shape, and reference agreement	Primary loss, residual multiples, transform artifacts, and amplitude imbalance	12
Shot-to-CMP sorting (shot_to_cmp_sorting)	Reorganize shot-domain traces into CMP panels with valid geometry	Header mapping, ordering, and CMP assignment	Order and shape validity, sample conservation, fold coverage, and metadata	Trace loss or duplication, incorrect ordering, sparse fold, and axis mismatch	8
Velocity-model construction (velocity_model_building)	Build or smooth a physically plausible migration velocity model	Interpolation, smoothing, physical constraints, and model parameterization	Physical range, profile error, roughness, structure, and reference agreement	Nonphysical velocities, discontinuities, over-smoothing, and structure loss	10
Zero-offset reverse-time migration (zero_offset_rtm)	Construct a focused zero-offset image from data and a velocity model	Velocity input, forward-modeling or migration settings, boundary handling, and image selection	Image focus, edge artifacts, positioning, amplitude stability, shape, and reference agreement	Boundary reflections, defocus, numerical instability, and incorrect image geometry	12

C Environment, Tools, and Artifact State

C.1 Latent state and policy projection

The latent environment state is

$$z_t = (\mathcal{A}_t, \mathcal{G}_t, h_t, \hat{x}_t, m_t, b_t, y^\star), \tag{13}$$

where $\mathcal{A}_t$ is the artifact set, $\mathcal{G}_t$ is the lineage graph, h_t is the complete action/result history, $\hat{x}_t$ is the visibly supported best candidate, m_t stores evaluator and workflow counters, b_t is the remaining budget, and $y^\star$ is private verifier state. The policy receives only the compact projection

$$o_t = (g, s(\mathcal{A}_t), \widetilde{h}_t, \mathcal{F}_t, \widehat{m}_t, b_t), \tag{14}$$

where $s(\mathcal{A}_t)$ contains visible summaries, $\widetilde{h}_t$ is bounded history, $\mathcal{F}_t$ is the state-valid interface set, and $\widehat{m}_t$ contains visible candidates, reference-free diagnostics, contract progress, and finish blockers.

Each successful call creates a persistent artifact record containing the artifact path, semantic role, parent artifacts, public interface, resolved parameters, creation step, compact summary, and execution status. Later calls may inspect or transform any valid artifact. This supports branching, candidate comparison, return to earlier states, and verification that the selected output is derived from task inputs.

C.2 Action format

The policy emits exactly one schema-constrained function call and no separate free-form rationale. A processing call has the form

```
<tool_call>
{"name":"nmo_apply","arguments":{
  "input_rsf":"input_00.rsf",
  "velocity":2200,
  "output_rsf":"candidate_001.rsf"}}
</tool_call>
```

and termination has the form

```
<tool_call>
{"name":"finish","arguments":{
  "answer":"Selected the best evaluated candidate.",
  "selected_output":"candidate_001.rsf"}}
</tool_call>
```

Malformed wrappers, invalid JSON, multiple calls, unavailable interfaces, missing or mistyped parameters, failed preconditions, paths outside the episode workspace, input overwrites, and invalid output destinations are rejected and recorded. Failed calls consume interaction budget and cannot silently alter the workspace.

C.3 Policy-Visible Prompt and Interaction Example

Figure 4 gives an abridged example from a real spiking-deconvolution rollout. It illustrates the policy-visible instruction, artifact aliases, available operations, reference-free candidate evidence, and workflow progress. Repeated state fields and full JSON schemas are omitted for readability; the complete prompt, schemas, and trajectory will be included in the public release.

The complete episode contains six evaluated candidates, five post-QC revisions, and 18 meaningful operation calls, with no invalid action or operation failure. Candidate selection uses visible deconvolution diagnostics. After termination, the evaluator reports SSIM 0.6755, input-relative SSIM gain 0.2575, and correlation 0.8790.

C.4 Environment loop

At every turn, the environment renders the compact reference-free state, validates exactly one policy call, records any failure, and updates artifacts, lineage, candidate evidence, workflow counters,

Task instruction. Compress the source wavelet and improve temporal resolution while preserving event timing and avoiding unstable whitening. Compare materially different candidates and retain the best-supported processed RSF.

Policy-visible state. Input: `input_00.rsf`; no accepted target or reference score is visible; 28 steps remain; no candidate has yet been evaluated.

Available operations (excerpt). `wiener_decon`, `bandpass_filter`, `evaluate_decon_candidate`, and `finish`.

Initial candidate and visible evidence. `wiener_decon(pnoise=0.01)` followed by `bandpass_filter(5--55 Hz)` produces a candidate with visible score 0.7566, low-frequency leakage 0.0626, wavelet width 10.6 ms, and RMS ratio 0.339.

Evidence-driven revision. Regularizer-only changes plateau near 0.760. Revising both prewhitening and cleanup bandwidth produces scores of 0.8456 at `pnoise=0.20, 8--50 Hz` and 0.8818 at `pnoise=0.30, 8--50 Hz`, with leakage reduced to 0.0011.

Supported finish. The policy selects the explicitly evaluated artifact `candidate_012.rsf`. SSIM, SSIM gain, and correlation are computed only after termination.

Figure 4: **Abridged policy-visible prompt and evidence-driven revision.** The full context additionally contains typed operation schemas, bounded history, best-so-far state, finish blockers, and remaining budget. No accepted target, reference path, or reference-derived metric is visible to the policy. This is a training/development rollout used to illustrate the interaction protocol, not held-out test evidence.

blockers, and remaining budget. Algorithm 1 gives the complete transition logic used by both training and evaluation.

C.5 Operation registry and backends

The registry exposes 88 typed public interface names. They map to 68 semantic groups: 52 scientific processing or modeling groups, seven inspection groups, eight evaluation groups, and one artifact-management/finalization group. The public count includes aliases and backend alternatives and is not a claim of 88 independent seismic algorithms. The audited mapping contains 67 Python-handler interfaces and 21 Madagascar-wrapper interfaces; another view classifies 56 names as distinct semantic interfaces, 21 as wrapper/backend alternatives, and 11 as aliases or shared handlers.

Table 8: Operation-registry views. Aliases and alternative backends explain why the public-name count is larger than the scientific-group count.

Registry view	Count
Typed public names	88
Scientific processing/modeling groups	52
Inspection groups	7
Evaluation groups	8
Artifact-management/finalization groups	1
Python-handler interfaces	67
Madagascar-wrapper interfaces	21

Calls follow the path

$$\text{policy call} \rightarrow \text{schema/allowlist} \rightarrow \text{handler dispatch} \rightarrow \text{result} \rightarrow \text{lineage}. \quad (15)$$

An alias invokes a shared handler, while Python and Madagascar variants are treated as different implementations. They are not assumed to be numerically identical because interpolation, boundary treatment, normalization, and metadata handling can differ. The trajectory records the public interface and arguments, and the release logs record the resolved handler and software version.

Algorithm 1 Executable seismic interaction loop

Copy task inputs to neutral workspace aliases.
Initialize artifacts, lineage, history, counters, blockers, and budget.
for $t = 0, \ldots, B_\tau - 1$ **do**
    Render a reference-free observation and state-valid schemas.
    Parse exactly one processing or finish call.
    if the call is invalid **then**
        Record the failure and return compact feedback.
    else if the call is finish **then**
        Compute public artifact and workflow blockers.
        if no blocker remains **then**
            Persist the selected artifact and mark the episode complete.
        else
            Return the blockers and continue.
        end if
    else
        Execute the typed operation through the local dispatcher.
        Register outputs, lineage, summaries, candidates, and evaluations.
        Update visible best state, revision counters, and remaining budget.
    end if
    During training, compute private reward for the recorded action outcome, including invalid, failed, and finish actions.
    if the episode is complete **then**
        Terminate.
    end if
end for
If the budget is exhausted, record truncation.
After the episode, compute reference metrics and task-aware QC.

C.6 Dynamic operation masks

The policy does not choose among all 88 interfaces at every turn. Each family starts from a task-compatible allowlist, after which state preconditions remove calls that cannot yet execute. For example, a candidate evaluator becomes available only after a compatible artifact exists, and migration requires both data and model artifacts. This structure removes irrelevant failures but does not specify operation order, numerical parameters, artifact dependencies, revision, or stopping. The family-operation matrix distributed with the release records visible, state-dependent, contract-required, role, and evaluator interfaces.

Table 9: Representative interfaces documented by the implementation. The complete 88-name catalog, aliases, handlers, and family allowlists are emitted as machine-readable release tables; the listed interfaces show the main scientific coverage without counting backend variants as separate algorithms.

Interface group	Representative public names	Scientific role
RSF input/output and inspection	inspect_rsf, summarize_rsf, sfin, sfattr, read_rsf, write_rsf, plot_gather	Read headers and sidecars; summarize dimensions, physical axes, units, amplitudes, energy, and frequency; produce audit views
Velocity analysis and NMO	velocity_scan, velocity_scan_cube, pick_velocity_field, nmo_apply, nmo_apply_time_velocity, nmo_field_apply, nmo_time_velocity_mute_scan	Build semblance or velocity evidence, select constant or time-varying velocities, apply moveout correction, and control stretch/muting
Stacking and coherence	stack_cmp, stack_cmp_clean, txspf_denoise, txspf_denoise_scan	Combine corrected traces, measure or improve coherence, and create stacked products
Filtering and muting	fk_filter, fk_filter_scan, mute_top, apply_mutter, apply_mutter_scan, ground_roll_separation	Attenuate coherent noise, apply mute boundaries and tapers, scan filtering parameters, and separate ground roll
Deconvolution	wiener_decon, wiener_decon_scan, sfpef	Estimate or apply predictive/spiking filters and compare operator settings
Radon processing	hyperbolic_radon_fwd, hyperbolic_radon_inv, radon_conjgrad, radon_demultiple	Transform to and from Radon domains, solve inverse systems, and reconstruct primary-preserved outputs
Migration and wave propagation	prestack_migrate, fd_zerooffset_model, fd_rtm_migrate, sfmig2, sfmigration	Perform prestack or zero-offset imaging and related wave-propagation operations
Velocity and density models	smooth_velocity, build_density_from_velocity_rsf, reflectivity_from_velocity	Smooth or construct physical models and derive auxiliary reflectivity products
Geometry conversion	read_segy, bin_segy_geometry, shot2cmp, sfshot2cmp	Read trace headers, bin irregular geometry, and reorganize shot-domain traces into CMP products
Evaluation and finalization	compute_snr, compute_flatness_metrics, copy_best_to_target, finish	Compute reference or reference-free diagnostics, register candidate evidence, and select a supported final artifact

D Task Contracts and Finish Conditions

D.1 Basic artifact contract

A finish action is legal only when all applicable conditions pass:

1. the RSF header and binary sidecar, or corresponding SEG-Y-derived product, exist and are readable;
2. values are finite, nonblank, and nontrivial;
3. shape, axes, units, and metadata satisfy the requested output role;
4. the output is confined to the episode workspace;
5. lineage traces the selected artifact to one or more task inputs;
6. the output is not an unchanged input copy;
7. a transformation task does not return an analysis-only product;
8. the workflow invokes the required scientific operation group; and
9. the selected artifact has the evidence required by the task contract.

These checks distinguish creation of a path from construction of a valid scientific product. File safety, finite-content checks, lineage, and workspace confinement remain active in all conditions.

D.2 Iterative candidate–assess–revise contract

All 45 iterative tasks require at least three distinct processing candidates, three explicit candidate evaluations, three material parameter variants, two revisions after observed evaluation, and selection of an evaluated final candidate. A candidate is a successful processing output with valid lineage and a processing role. Candidates are materially different when their scientific parameter signatures or relevant upstream lineage differ after output-path arguments are removed. An evaluation is an explicit call to the configured reference-free evaluator after candidate creation. A revision is a later material processing decision made after evaluation of an earlier candidate. The contract verifies that

candidate evaluation precedes material revision. Revision counts describe this observable interaction pattern rather than identify the causal effect of feedback on the policy.

Visible counters and finish blockers tell the policy which public requirements remain unsatisfied. This makes the protocol reproducible across backbones and prevents a task from being marked complete by an unevaluated file. Revision counts are therefore treated as descriptive workflow measures rather than as an unconditional quality metric.

E Scientific Evaluation Details

E.1 Evaluation record

For final artifact $\hat{x}$ and trajectory ξ, the evaluator returns

$$\mathbf{M}_\tau(\hat{x}, \xi) = (V_\tau, \mathbf{d}_\tau, \mathbf{p}_\tau, \mathbf{c}_\tau)\,, \tag{16}$$

where V_τ is valid completion, $\mathbf{d}_\tau$ stores scientific and reference-agreement measurements, $\mathbf{p}_\tau$ stores artifact or preservation failures, and $\mathbf{c}_\tau$ stores workflow behavior and cost. The vector form is retained because physical meaning differs across seismic products.

E.2 Reference-agreement measures

For aligned arrays x and y, the evaluator reports

$$\mathrm{Corr}(x, y) = \frac{\langle x - \bar{x}, y - \bar{y} \rangle}{\|x - \bar{x}\|_2 \|y - \bar{y}\|_2 + \epsilon}, \tag{17}$$

$$\mathrm{NRMSE}(x, y) = \frac{\|x - y\|_2}{\|y\|_2 + \epsilon}, \tag{18}$$

$$\mathrm{PSNR}(x, y) = 20 \log_{10} \frac{R}{\sqrt{\mathrm{MSE}(x, y)} + \epsilon}, \tag{19}$$

where R is the configured data range. SSIM, output-to-target amplitude ratio, shape validity, and gains relative to the unprocessed input are also recorded. These metrics compare with one accepted target; family-specific diagnostics remain necessary because multiple seismic outputs can be scientifically acceptable.

The automatic reference composite used by the main experiment is

$$\begin{aligned} q_{\mathrm{ref}} = {} & 0.28 q_{\mathrm{corr}} + 0.22 q_{\mathrm{nrmse}} + 0.16 q_{\mathrm{ssim}} \\ & + 0.09 q_{\mathrm{psnr}} + 0.08 q_{\mathrm{amp}} + 0.06 q_{\Delta\mathrm{psnr}} \\ & + 0.04 q_{\Delta\mathrm{ssim}} + 0.02 q_{\Delta\mathrm{lpips}} + 0.05 q_{\mathrm{shape}}, \end{aligned} \tag{20}$$

with components normalized and clipped to $[0, 1]$. Missing components receive the neutral value 0.5. Learned perceptual image patch similarity (LPIPS) was disabled in the canonical run, so it is not reported as a primary endpoint. When input-relative gains are unavailable, the implementation uses the corresponding fixed no-gain branch documented in the metric configuration.

The failure-aware task score and aggregates are

$$q_\tau = V_\tau \, \mathrm{clip}(q_{\mathrm{ref},\tau}, 0, 1), \tag{21}$$

$$Q_{\mathrm{sci}} = \frac{1}{19} \sum_{\tau \in \mathcal{T}_{\mathrm{test}}} q_\tau, \tag{22}$$

$$Q_{\mathrm{iter}} = \frac{1}{9} \sum_{\tau \in \mathcal{T}_{\mathrm{iter,test}}} q_\tau. \tag{23}$$

Because the test split contains one task per family, these equations are equal family macros. Invalid tasks receive zero instead of being removed.

E.3 Compatible input-relative SSIM

Input-relative structural improvement is

$$\Delta\mathrm{SSIM} = \mathrm{SSIM}(\hat{x}, y^\star) - \mathrm{SSIM}(x_{\mathrm{input}}, y^\star). \tag{24}$$

It is reported only for the 11 families whose input and output products permit meaningful aligned comparison: automatic gain control, coherence enhancement, datum statics, spatially varying NMO, first-break mute, F–K filtering, ground-roll suppression, CMP NMO, Radon demultiple, spiking deconvolution, and stacking-velocity analysis. Every aggregate cell reports its valid denominator.

E.4 Artifact warnings and workflow measures

Let $p_{\tau,j} \in \{0, 1\}$ denote a prespecified artifact or preservation failure. The task-level warning indicator is

$$w_\tau = \mathbf{1}\left[V_\tau = 0 \ \vee \ \max_j p_{\tau,j} = 1\right], \tag{25}$$

and the warning rate for one deterministic 19-task evaluation is

$$\mathrm{WarnRate} = \frac{100}{19} \sum_{\tau=1}^{19} w_\tau. \tag{26}$$

A task with several warning conditions is counted once. For trainable methods, the rate is first computed within each seed and then summarized across the three seeds.

Meaningful operations count successful nontrivial inspection, processing, or evaluation calls that advance the artifact workflow. Post-QC revisions count material processing changes made after an earlier candidate evaluation. Total calls, candidates, evaluations, parameter variants, invalid actions, blocked finishes, tool failures, budget truncation, generated tokens, and runtime are retained in the detailed reports. These workflow measures have no universal better direction and are interpreted jointly with validity and artifact quality.

E.5 Automatic QC status

The deterministic warning layer checks missing required metrics, shape mismatch, correlation below 0.65, NRMSE above 0.55, amplitude ratio below 0.45 or above 2.25, PSNR gain below -0.5 dB, SSIM gain below -0.03, LPIPS reduction below -0.02 when enabled, composite score below 0.45, or visual penalty above 0.35. Status is assigned as follows:

- ok: operational completion with no configured warning;
- review: operational completion with one or more warnings;
- fail: operational failure.

These statuses are deterministic engineering indicators. The family-specific measurements and task-aware panels provide the scientific interpretation of a warning.

E.6 Family-specific scientific evidence

Representative dedicated candidate scores are

$$q_{\mathrm{NMO}} = (0.70 q_{\mathrm{flat}} + 0.30 q_{\mathrm{coh}}) g_{\mathrm{preserve}} g_{\mathrm{stretch}}, \tag{27}$$

$$q_{\mathrm{FK}} = 0.55 \max(0, q_{\text{low-velocity reduction}}) + 0.45 g_{\mathrm{preserve}} g_{\mathrm{stability}}, \tag{28}$$

$$q_{\mathrm{GR}} = \mathrm{clip}\left(\sqrt{q_{\mathrm{suppression}} q_{\mathrm{band}}}(0.85 + 0.15 s), 0, 1\right), \tag{29}$$

where the g terms are preservation guards and s is transition softness. The field-NMO evaluator uses weights 0.55/0.15/0.10/0.10/0.05/0.05 for stack coherence, adjacent coherence, amplitude, coverage, entropy, and active support before collapse and event guards. The deconvolution evaluator uses weights 0.16/0.14/0.12/0.18/0.08/0.05/0.20/0.02/0.05 for phase, width, compactness, low-frequency guard, sidelobes, sharpness, amplitude, crest factor, and coherence. The generic observable fallback is

$$q_{\mathrm{obs}} = 0.70 q_{\mathrm{flatness}} + 0.30 q_{\text{stack coherence}}, \tag{30}$$

followed by nonzero-support, root-mean-square, energy-balance, and artifact guards. Raw physical measurements are retained for interpretation and are not compared directly across unrelated families.

Table 10: Scientific evidence used across the 19 families. Closely related families share a row when they use the same scientific interpretation.

Family	Primary scientific evidence	Online/offline use
Automatic gain control	Robust-amplitude balance, gain behavior, adjacent-trace correlation, singular-value energy, active/quiet contrast, and quiet fraction	Dedicated reference-free evaluator plus offline reference
Coherence enhancement	Flatness, stack coherence, adjacent correlation, and trace-energy balance	Conservative reference-free diagnostics plus offline reference
Datum statics	Event alignment and static-shift profile	Structural and offline reference evaluation
Density model	Physical density range and velocity-density relation residual	Structural checks plus correlation/NRMSE/SSIM
Spatially varying NMO and stacking velocity	Stack and adjacent coherence, amplitude, coverage, entropy, active support, residual moveout, event continuity, pick smoothness, and collapse/stretch guards	Dedicated field-NMO reference-free evaluator plus offline reference
First-break mute	Early-energy attenuation, live fraction, coherence, event support, and preservation	Conservative reference-free diagnostics plus offline reference
F–K filtering	Low-apparent-velocity attenuation, reflection/pass-band preservation, stability, and ringing guards	Dedicated reference-free evaluator plus offline reference
F–K spectrum	Axis/shape validity, spectral-energy distribution, and target agreement	Structural/offline evaluation
Geometry binning	Trace/sample conservation, fold range, filled-bin fraction, geometry order, axes, and metadata	Structural/offline evaluation
Ground-roll suppression	Low-velocity attenuation, signal-band preservation, transition stability, ringing, and amplitude guards	Dedicated reference-free evaluator plus offline reference
CMP NMO	Event flatness, stack coherence, offset support, amplitude preservation, and stretch/mute guards	Dedicated reference-free evaluator plus offline reference
NMO stack	Coherence, continuity, retained variance, shape, and target agreement	Structural/offline evaluation
Prestack migration and RTM	Image focus, edge artifacts, positioning, amplitude stability, shape, and target agreement	Structural/offline evaluation
Radon demultiple	Multiple attenuation, primary preservation, component balance, shape, and target agreement	Structural/offline evaluation
Shot-to-CMP sorting	Order/shape validity, sample conservation, fold coverage, and metadata	Structural/offline evaluation
Spiking deconvolution	Wavelet phase and width, compactness, low-frequency guard, sidelobes, sharpness, amplitude, crest factor, and coherence	Dedicated reference-free evaluator plus offline reference
Velocity model	Physical range, profile error, roughness, structure, and target agreement	Structural/offline evaluation

E.7 TASK-AWARE RENDERING

Offline evaluation routes each family to a product-aware renderer. Gather families show input, accepted target when available, final output, and residual or task-specific differences under robust shared clipping. Velocity and NMO panels show scans, picks, corrected gathers, and residual moveout. Statics panels show alignment and shift profiles; geometry panels show trace maps, fold, and filled bins; F–K panels show spectral axes and energy support; model panels show physical grids and profiles; Radon panels show transformed components and reconstruction; migration panels show image focus, positioning, and edge behavior. Rendered pixels do not enter the policy observation and are not used directly as reward.

Figure 5 shows four locked-test examples under a fixed comparison layout. Each row uses the same column semantics: task input, agent-selected output, and accepted reference. The examples span spatially varying NMO correction, first-break muting, F–K filtering, and spiking deconvolution, covering distinct processing objectives and preservation requirements.

The rows illustrate complementary scientific criteria. Field NMO emphasizes event flattening with controlled stretch; first-break muting balances early-energy removal against retained reflection support; F–K filtering suppresses coherent dipping noise while preserving the reflection band; and spiking deconvolution sharpens temporal events while controlling leakage and energy distortion. Complete task-aware panels, residual views, and numerical diagnostics for all families will be included in the public release.

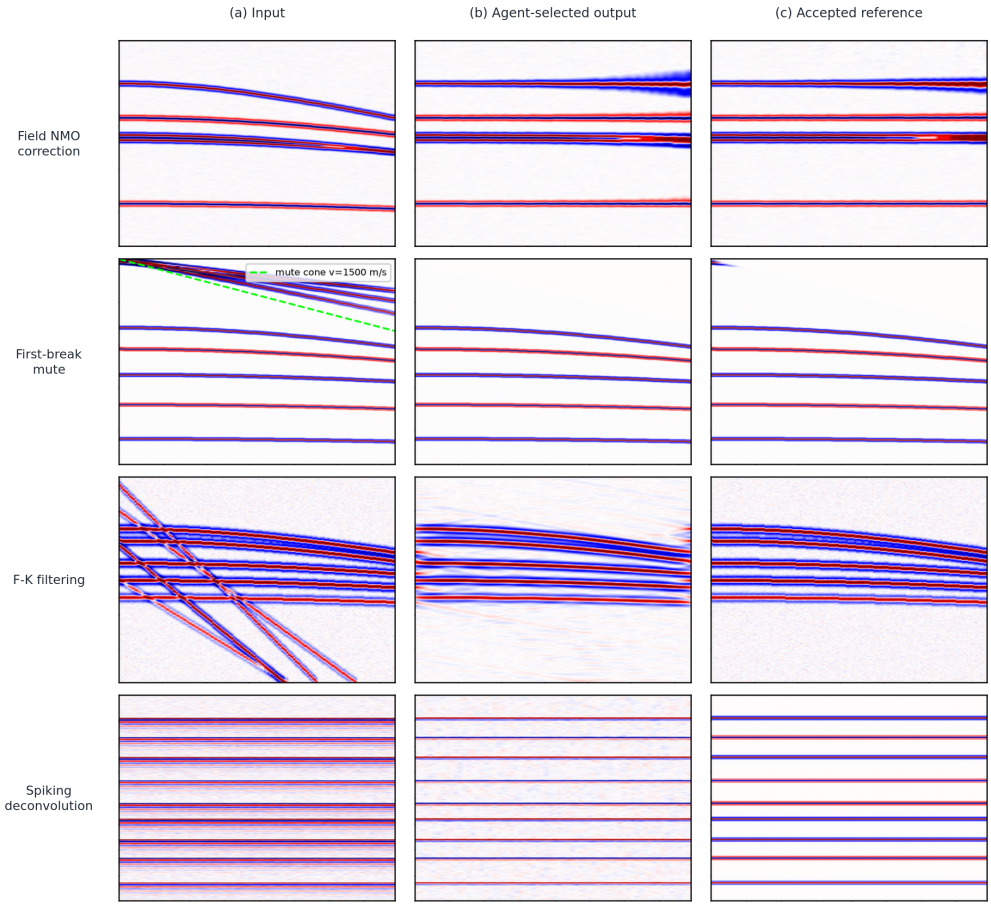


Figure 5: **Representative cross-family processing results.** Columns show the task input, agent-selected output, and accepted reference with identical semantics in every row. Accepted references are hidden during interaction and are displayed only for post-episode evaluation.

F SeisAgent Training Details

F.1 Executable reward

The main paper groups the step reward as

$$r_t = r_t^{\text{exec}} + r_{\tau,t}^{\text{quality}} - \lambda_p p_{\tau,t}^{\text{art}} - \lambda_c c_t. \tag{31}$$

The implementation expands this form into call validity, operation success, observable candidate improvement, private reference improvement, new-best bonuses, preservation guards, step cost, finish support, and terminal validity. Missing outputs, malformed actions, and failed execution are handled by r_t^{exec}; $p_{\tau,t}^{\text{art}}$ applies to produced candidates that pass basic execution checks but damage signal or structure.

Reference-derived reward terms remain private. The policy receives execution status and permitted reference-free diagnostics, but not the scalar reward, accepted artifact, reference path, or reward breakdown. At test time, clean references are used only after the episode.

Table 11: Resolved reward coefficients used by the paper protocol. Values are magnitudes; penalties enter with negative sign.

Configuration field	Value	Role
step_penalty	0.010	Cost for consuming an environment step
tool_call_penalty	0.025	Cost for executing an operation call
late_step_penalty	0.020	Additional cost scaled by relative episode position
success_bonus / failure_penalty	0.040 / 0.250	Immediate execution outcome
invalid_penalty / repeated_failed_penalty	0.450 / 0.200	Malformed, unavailable, or repeatedly failing calls
snr_improvement_scale_db / weight	6.0 / 0.900	Normalized hidden reference-SNR improvement
snr_new_best_bonus / minimum gain	0.250 / 0.250 dB	Reward for a new best candidate under the SNR verifier
snr_first_measurement_bonus	0.120	First usable hidden measurement
quality_improvement_scale / weight	0.100 / 0.350	Improvement in observable task-specific candidate score
quality_new_best_bonus / first-measurement weight	0.080 / 0.080	New-best and first observable candidate signals
supervised_quality_weight	0.250	Private normalized reference-agreement component during intermediate processing
low_nonzero_penalty	0.250	Collapse or near-blank support
low_reference_similarity_penalty	0.350	Very weak private reference agreement
shape_mismatch_penalty	0.700	Invalid output geometry or shape
residual_artifact_penalty	0.400	Residual or collateral artifact condition
final_snr_scale_db / weight	8.0 / 1.000	Final hidden SNR term
final_below_best_penalty	0.100	Selecting a candidate below the best recorded verifier score
final_quality_weight / final_supervised_quality_weight	0.250 / 0.450	Final observable and private quality terms
final_best_bonus	0.150	Selecting the recorded best candidate
final_success_bonus / final_failure_penalty	0.300 / 1.250	Finish-level task outcome
final_budget_bonus	0.200	Bounded benefit for unused budget after supported completion
terminal_success_bonus / terminal_failure_penalty	0.500 / 1.500	Episode terminal outcome
terminal_missing_output_penalty	2.500	Termination without a final artifact

F.2 Group-relative executable policy optimization

For each sampled task, the policy generates $G = 4$ complete trajectories. The discounted return and same-task advantage are

$$R_i = \sum_{t=0}^{T_i} \gamma^t r_{i,t}, \qquad \gamma = 0.99, \tag{32}$$

$$A_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}. \tag{33}$$

When process credit is enabled, transition credit is

$$A_{i,t} = A_i + \lambda_{\text{proc}} \frac{r_{i,t} - \mu_{\text{step}}}{\sigma_{\text{step}} + \epsilon}, \qquad \lambda_{\text{proc}} = 0.50. \tag{34}$$

If group return variation does not exceed the configured standard-deviation floor of 0.10, the episode-level advantages are set to zero. Transition credit can still be used when immediate rewards vary; otherwise the update is skipped.

Let $\ell^{\theta}_{i,t}$ and $\ell^{\text{old}}_{i,t}$ be the summed log probabilities of generated action tokens under the updated and rollout policies, and define

$$\rho_{i,t}(\theta) = \exp(\ell^{\theta}_{i,t} - \ell^{\text{old}}_{i,t}). \tag{35}$$

For N valid transitions, the implementation minimizes

$$\mathcal{L}_{\text{GR}}(\theta) = \frac{1}{N} \sum_{i,t} \Big[-\min\big(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon) A_{i,t}\big) + \beta(\ell^{\text{old}}_{i,t} - \ell^{\theta}_{i,t})\,\text{sg}(\rho_{i,t}) \Big],$$

We use $\varepsilon = 0.15$ and $\beta = 0.04$. The second term is an importance-weighted auxiliary loss on sampled actions, with sg stopping gradients through the importance weight; it is not a KL-divergence penalty. The group-relative update uses likelihood-ratio clipping and no learned value model or separately loaded reference model. Gradients are accumulated over the retained transitions and normalized by their count.

Table 12: Group-RL exposure and optimization settings.

Group-RL setting	Value
Unique RL task pool	55
Epochs	5
Task groups per epoch	11
Rollouts per task group	4
Rollouts per epoch	44
Total task groups / trajectories	55 / 220
Task sampling	Family-interleaved, no replacement
Rollout temperature	0.9
Discount γ	0.99
Minimum group return std.	0.10
Process-credit coefficient	0.50
Ratio clip ε	0.15
Sampled-action auxiliary coefficient β	0.04
Learning rate	1.5×10^{-6}
Gradient norm cap	1.0
Adaptive extra rollouts	Disabled
Value model / frozen reference LLM	None / none

G Experimental Protocol and Reproducibility

G.1 Paper-facing experimental matrix

The backbone block evaluates Qwen3-4B, Qwen3-8B, Qwen3-14B, Qwen3-32B, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 under zero-shot interaction, two-example schema in-context learning (ICL-2), and one frozen SFT-2+Group-RL recipe. The training-configuration block on Qwen3-14B adds SFT-1, SFT-2, Group-RL only, and SFT-1+Group-RL; the SFT-2+Group-RL row is shared with the backbone block. Two deterministic scripts test a family-default workflow and a reference-visible fixed-script sensitivity control.

The unique paper matrix contains 30 trainable seed runs and 14 deterministic evaluation conditions. At 19 test tasks per condition or seed, the locked test study contains 836 task episodes. Eight unique Group-RL condition types are run with three seeds and 220 trajectories per seed, corresponding to 5,280 complete Group-RL training trajectories. Every saved SFT or Group-RL epoch is evaluated on the 19-task development split before checkpoint locking.

G.2 Backbone and adaptation settings

All trainable conditions use bfloat16 low-rank adaptation (LoRA), no quantization, rank 32, scaling factor 64, dropout 0.05, no LoRA bias, and adapters on all linear layers. AdamW is used with gradient clipping at 1.0; gradient checkpointing is enabled and the key-value cache is disabled during training. For Qwen3-14B, this configuration trains 128,450,560 of 14,896,757,760 parameters, or 0.8623%.

G.3 Hardware and software

Table 13: Compute node used for the paper experiments.

Component	Configuration
Accelerators	4× NVIDIA A100-SXM4 80 GB
GPU memory	81,920 MiB per GPU
CPU	Intel Xeon 2.20 GHz, 24 physical / 48 logical cores
Host memory	Approximately 668 GiB
Driver	NVIDIA 595.71.05
OS kernel	Linux 6.12 series

Table 14: Audited software environment.

Package	Version
Python	3.12.13
PyTorch	2.11.0+cu128
Transformers	5.13.0
PEFT	0.19.1
Accelerate	1.14.0
bitsandbytes	0.49.2
NumPy	2.4.4
SciPy	1.18.0
scikit-image	0.26.0
Matplotlib	3.11.0
pandas	3.0.3
PyYAML	6.0.3
safetensors	0.8.0

The operation interface is RSF/Madagascar-compatible and also provides pure-Python implementations. Native Madagascar commands were not visible on the audited cloud PATH; the release therefore records the actual resolved backend rather than assuming that every public wrapper executed a native binary.

H Additional Domain Analysis

H.1 Additional full CMP NMO trajectory

The main paper presents a compact NMO illustration. A longer recorded episode provides a complete example of inspection, parameter revision, and supported termination. The policy operates under reference-free feedback; final SNR and SSIM are computed only after the episode.

Table 15: A complete CMP NMO trajectory recorded during development. Offline metrics for the selected artifact were SNR 5.809 dB and SSIM 0.874; neither was visible to the policy.

Step	Operation	Evidence or state change
1	Inspect a 500×21 CMP gather	Record axes, amplitudes, and event structure
2	Velocity scan	Strong evidence near 2708.9 m/s at 1.524 s
3	Apply constant-velocity NMO	Create the first corrected candidate
4–5	Evaluate and confirm flatness	Flatness/coherence 0.481/0.370
6	Compute windowed semblance picks	Build time-varying velocity evidence
7	Apply time-varying NMO	Create a revised candidate
8–9	Evaluate and confirm	Flatness/coherence 0.627/0.545
10	Revise semblance smoothing	Change the velocity-field regularization
11	Reapply time-varying NMO	Create a new candidate
12–13	Evaluate and confirm	Flatness/coherence 0.669/0.708
14	Compute event-constrained picks	Use event support to revise velocity
15	Apply event-aware NMO	Create another candidate branch
16–17	Evaluate and confirm	Flatness/coherence 0.719/0.690
18	Revise the event window	Adjust the event-support region
19	Apply stretch mute	Maximum stretch set to 0.3
20	Evaluate the selected candidate	Flatness/coherence 0.686/0.772
21	Finish with candidate_008.rsf	Selected artifact copied to the task output after contract checks

H.2 Scope

The current benchmark uses five tasks per family and controlled, project-assembled artifacts. It does not establish deployment on independent field surveys or unseen processing families. Family-compatible operation masks and finish contracts make execution reproducible and reduce irrelevant calls; they also provide scientific structure. Accepted targets and automatic metrics support scalable comparison but do not replace blinded expert assessment. These scope limits do not change the main purpose of the benchmark: to provide a shared executable setting in which policies must construct, inspect, revise, and select seismic artifacts under domain-specific evidence.